

2018A F=ma Exam: Problem 2

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The velocity of the center of mass is given by

$$\vec{v}_{CM} = \frac{\vec{p}_{tot}}{m_{tot}}$$

Since momentum is conserved, we can compute $\vec{v}_{CM}$ before the collision. Thus,

$$\vec{v}_{CM} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} = \frac{(3.0 \text{ kg})(30 \text{ m/s}) + (2.0 \text{ kg})(-2.0 \text{ m/s})}{3.0 \text{ kg} + 2.0 \text{ kg}} = 10 \text{ m/s}$$

so the answer is B.